O-H-205D
March 12, 1976
SUPERSEDING
O-H-205C
September 22, 1969

FEDERAL SPECIFICATION

HERBICIDE, DALAPON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a water soluble, powdered herbicide made up of sodium salts or a combination of sodium and magnesium salts of dalapon (2,2-dichloropropionic acid) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-D-723 - Drums, Fiber.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6840

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Laws and Regulations:

40 CFR 162-130 - Pesticide Programs.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition. The dalapon herbicide shall be composed of active and inert ingredients, as specified herein. The product shall be a free-flowing powder which shall be dry, loose, and free from foreign matter.

3.1.1 Active ingredient. The active ingredient shall be the sodium salt or a combination of sodium and magnesium salts of 2,2-dichloropropionic acid. The acid equivalent shall be 74.0 percent by weight, when tested as specified in 4.4.2.1.

3.2 Solubility. 120 grams of the herbicide shall be completely soluble in one liter of water (1 pound per gallon) at room temperature (23° ± 3°C), when tested as specified in 4.4.2.2.

3.3 Registration and labeling. The herbicide shall be registered and labeled in compliance with the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), as amended. This act, and regulations promulgated thereunder, is administered by the Environmental Protection Agency (EPA).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Inspection sampling. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105, at the inspection levels and acceptable quality levels (AQL's) specified herein.

4.3 Examination.

4.3.1 Examination of preparation for delivery.

4.3.1.1 Sampling plan. The lot size shall be the number of containers offered for inspection at one time. The sample unit shall be the contents of one filled shipping container. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

4.3.1.2 Examination procedure. An examination shall be performed to determine compliance with the requirements of section 5. The sample units shall be examined for the defects described in Table I.

TABLE I

Examine	Defect
Contents	Not as specified.
Containers	Not as specified.
Packing	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Bulging or distortion of container. Tears, cracks, splits, holes, or cuts; closure incomplete; leakage or seepage from any part of the container. Cushioning inadequate, improper, or missing.

4.3.2 Examination of contents of unit container.

4.3.2.1 Sampling plan. A lot shall consist of all of the containers filled with the herbicide of one type produced by one manufacturer under the same conditions and process, within a 24 hour period. The sample unit shall be one such filled container. The sample shall be selected in accordance with level S-2 and the AQL shall be 4.0 percent defective.

4.3.2.2 Examination procedure. The examination shall be made to determine compliance with section 3.1. The sample units shall be examined for defects and listed in Table II.

TABLE II

Examine	Defect
Materials	Form not as specified.
Workmanship	Foreign matter visible present; not free-flowing.

4.4 Testing.

4.4.1 Sampling for testing. The herbicide shall be tested for conformance to the requirements of section 3 in accordance with the test methods specified in 4.4.2. The lot size shall be all the herbicide of one type produced by the manufacturer in a 24 hour period, under the same conditions, using the same raw materials. The sample unit shall be a 1-pound composite, selected in equal portions from a lot in accordance with inspection level S-1. When the herbicide is sampled from a continuous-run process, the composite sample shall be made up of three 1/3 pound portions so as to represent, respectively the first part, the middle part, and the last part of the run which constitutes the inspection lot. The one pound composite shall be thoroughly mixed to yield the composite sample and stored in a clean, dry, glass container with the identifying lot number and date. All tests shall be performed on the composite sample. Care shall be exercised to prevent loss or absorption of moisture during sampling, weighing, and storage. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

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4.4.2 Test methods.

4.4.2.1 Acid equivalent of 2,2-dichloropropionic acid.

4.4.2.1.1 Precautions. The herbicide can cause mild skin and eye irritation. Avoid skin and eye contact. Carbon disulfide is toxic and flammable. Handle carefully in a fume hood protected from any source of ignition. Handle the sulfuric acid with caution and as recommended. The herbicide is a hygroscopic material. Exposure to the atmosphere must be kept to a minimum. Keep sample containers tightly closed before and after removal of sample.

4.4.2.1.2 Principle. The salt(s) of dalapon is (are) converted to 2,2-dichloropropionic acid which is extracted into carbon disulfide. The infrared absorption spectrum of the extract is qualitatively compared with a standard of the acid to confirm the identity of the product, and quantitatively compared with the standard at a wavelength which is specific for 2,2-dichloropropionic acid to determine the acid equivalent.

4.4.2.1.3 Apparatus.

(a) Infrared spectrometer.
(b) Cell, infrared 1.0 mm.

4.4.2.1.4 Reagents.

(a) Carbon disulfide, infrared grade.
(b) Sulfuric acid, 25 percent v/v. With caution, transfer 25 ml of concentrated sulfuric acid to 75 ml of distilled water.
(c) 2,2-dichloropropionic acid, 99 percent plus purity; available from the Sample Coordinator, Ag-Organics Dept., The Dow Chemical Company, Midland, Michigan 48640.
(d) Sodium chloride, reagent grade.

4.4.2.1.5 Calibration.

(a) Weigh accurately to the nearest 0.5 mg, 370 mg of 2,2 dichloropropionic acid (see 4.4.2.1.4c) into a 60 ml (2-ounce) bottle. Pipet exactly 50.0 ml of carbon disulfide into the bottle, cap, and mix until solution is complete. (This method is standardized so that 370 mg of 2,2-dichloropropionic acid represents 74 percent acid equivalent.)
(b) Scan the solution from 7 to 16 μ in the 1.0 mm cell on the spectrometer.
(c) With the base line drawn from 8.6 to 9.8 μ, measure the absorbance of the solution at 8.9 μ.

4.4.2.1.6 Procedure.

(a) Weigh accurately to the nearest 0.5 mg, 500 mg of sample into a 60 ml (2-ounce) bottle. Add 1 ml of 25 percent sulfuric acid and mix. Pipet 50.0 ml of carbon disulfide into the bottle, then add about 2 g of sodium chloride crystals. Shake for 30 minutes or until solution is complete.
(b) Scan the carbon disulfide solution from 7 to 16 μ in a 1.0 mm cell. Measure the absorbance as in 4.4.2.1.5.

4.4.2.1.7 Calculation.

(a) Acid equivalent, percent = $\frac{\text{sample absorbance}}{\text{standard absorbance}} \times \frac{\text{grams of standard}}{\text{grams of sample}} \times 100.$

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4.4.2.2 Solubility. Place 120 g of the herbicide in a clean, dry 1-liter glass-stoppered volumetric flask. Fill to the mark with distilled or deionized water. Shake vigorously until the herbicide dissolves. Allow the solution to stand for 1 hour. Examine for insoluble matter.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C (see 6.2).

5.1.1 Level A. The herbicide shall be packed in fifty-pound quantities in fiber drums conforming to PPP-D-723, Type II, Grade A.

5.1.2 Level B. The herbicide shall be packed in fifty-pound quantities in fiber drums conforming to PPP-D-723, Type I, Grade A.

5.1.3 Level C (Civil Agencies) and Commercial (Military Agencies). The herbicide shall be packed in fifty-pound quantities in fiber drums in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier. Containers and packing shall comply with Uniform Freight Classification or National Motor Freight Classification, and the Federal Insecticide, Fungicide, and Rodenticide Act.

5.2 Marking. Containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified.

5.3 Labeling. Containers shall be labeled in accordance with 40 CFR 162-180, Pesticide Programs (see 3.3).

6. NOTES

6.1 Intended use. Dalapon herbicides are recommended for control of annual and perennial grasses. Either alone or in combination, dalapon is suitable for land where long residual activity is not desirable.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Marking required (see 5.2).

MILITARY INTEREST

Custodians:
Navy - YD

Military Coordinating Activity:

Air Force - 68

Preparing Activity:
GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:
AGR-APS, ARS, AMS
GSA-FSS
HEW-FDA
VA-DMS

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